Ohm++

Improved Fishing Device Controller

A fishing rod that enables quadriplegics, paraplegics, and Traumatic Brain Injury patients to fish with an intuitive and easy to use controller.

Team: Garrett Hurd, Tyler Martin, Pedro Mejido, Vedant Shah,

Felip Tamayo-Tapia, Travis Wood

Mentors: John Lettang, Prof. Eric Bogatin

Sponsors: QL Plus – Scott Huyvaert, River Deep – Bob Adwar

Final Report

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Team Roles and Responsibilities

Team Member	Lead Administrative Roles	Backup Roles
Garrett Hurd	Project Lead, Schedule Manager	Logistics Manager, Assignments Manager
Tyler Martin	Customer Manager	Communications Manager, Morale Manager
Pedro Mejido	Scrum Master	Financial Manager
Vedant Shah	Financial Manager, Logistics Manager,	Project Lead
Felipe Tamayo-Tapia	Morale Manager	Schedule Manager
Travis Wood	Communications Manager, Assignments Manager	Scrum Master, Customer Manager

Team Member	Lead Engineering Roles	Backup Roles
Garrett Hurd	Power	System Architect, Analog Circuits
Tyler Martin	Analog Circuits, Testing, RF	Mechanical, Power
Pedro Mejido	System Architect, PCB	
Vedant Shah	Software,	User Interface, Embedded
Felipe Tamayo-Tapia	Embedded,	Software,
Travis Wood	User Interface, Mechanical	PCB, Testing, RF

Project Brief

"The What"

Quick Pitch:

We have created an intuitive, easy to use fishing rod controller for people who suffer from limited hand dexterity, spinal cord injuries, traumatic brain injuries, and incomplete/complete quadriplegics. This device will be used by patients at Craig Hospital during recreational fishing outing hosted by the River Deep Foundation. Our device works by taking in user inputs from our controller, those inputs then control our motor driver systems to imitate the user fishing, all this is power by 24V battery. Our number one priority was building the best user experience.

The Bigger Picture:

This project was a purpose driven endeavor. Our team was honored to go on a fishing trip with our sponsors where we were able to see the immediate impact that getting patients out of the hospital, in the outdoors, and actively participating in activities that they previously loved once again had on their morale. Being injured while in service, being in a car accident, or any other calamity that removes some of your own bodily autonomy can have far reaching consequences on your life, and often the inner turmoil that follows is ignored for the more obvious physical aspects of injury. Regaining control again to some degree can be uplifting and motivating, which is why our team fully subscribes to recreational therapy being a cornerstone of the healing journey.

When designing our project, we ensured that every engineering decision was filtered through the thoughts "How does this impact the end user experience" and "What does this accomplish to further our goal?" It is all too easy as engineers to come to a design decision that is cool, challenging, or impressive. It is much more difficult to keep the vision and mission of a project in sight, and align the vectors of individual team members in that direction.

Our device utilizes a user interface that had to go through multiple rounds of design in order to smooth out the details, develop around bodily ergonomics that are not in line with our own, and simplifying operations in order to ensure an easy learning experience that facilitates use by TBI patients. The entire project hinged on these design decisions, and we were adamant that we had to get it close to right early on in the process, so that we wouldn't be left with a device that worked from an electrical and computer engineering standpoint, but did nothing but collect dust in the Colorado summer because it couldn't be used by the people it was designed for.

Our project was a learning experience for all of us. More importantly it was a learning experience that had further reaching benefits than our own progression as engineers. We believe that a project like this further solidified our desires to better the world around us, and to continue our pursuit of knowledge and diverse experiences that that we may be better equipped to do even more of the same down the line.

Summary

Our sponsor had a small list of things that they wanted, and an even shorter list of things that they needed. Due to the target demographic for the device, safety was paramount. Because it would have to be carted out to the mountains, the return on investment for the effort required demands that the battery life exceed previous iterations of 5 minutes. Ideally a couple of hours. Lastly, the device absolutely had to be easy to use. The goal was that after using the device once with a guided walk through, that any patient would be up and running. Our metric for this in house was that if we give the controller to any passerby, that they could figure it out in under five minutes with no guidance other than what was printed on the controller itself

Needs:

- Safe – Doesn't catch on fire, get excessively hot and burn patients, or risk electrical shock
  - Met with ensuring that we minimized current draw, had no exposed wiring or connections, and that the electrical housing was rain proof.
- Battery Life of 2 Hours
  - Accomplished by ensuring adequate supply amp-hours, and by mitigating any parasitic losses, lowering idle power draw, and using only what was needed for individual motors with few to no states that had motors running unnecessarily, to include an idle state for the reeling motor.
  - The battery power lasted for 8 hours of operation with only a 25% reduction in power over those 8 hours of constant casting and reeling.
- Rain Resistance
  - Accomplished by using ABS 3D printed enclosures, vapor sealing that ABS and then sealing it further with spray on truck bed liner. All ventilation was designed to prevent rain splash intrusion into the device.
- Intuitive Controller
  - The controller was iterated with multiple sources to streamline the design. The controller only requires the in-place mounted labels to convey operation to a new user.

Wants:

- Lightweight Controller
  - ABS printed design reduced much of the weight that previous designs had per unit volume.
- Comfortable for extended use
  - The controller has a concave base to rest easily on a leg. A rubber pad was installed to add to the comfort
  - Straps have been installed on the bottom side to secure the controller to the leg of users
- Modular for future bite-tab integration
  - The controller connection is using a waterproof “Amphenol” style connection that can be easily removed and replaced with a future controller
- Easy to use for recreational therapists
  - The battery is attached to the side of the electrical housing, and is a drill battery that is easy to learn, and commonly known to many people. It also has an on-board battery power indicator

Technology Options

Power distribution:

We wanted to make it intuitive for users to replace the battery without having to remove any electrical systems. We opted to use a 24V 12.0 Ah lithium hand drill battery for the input voltage to the system. We placed this on the outside of the electrical housing for easy access. For the power distribution of the electrical system, we used a buck converter to down step the 24V input voltage to 12V. This 12V would be used to power our motor drivers and our motors. To power our microcontroller and our controller, we down stepped the 12V to 5V by using an LDO.

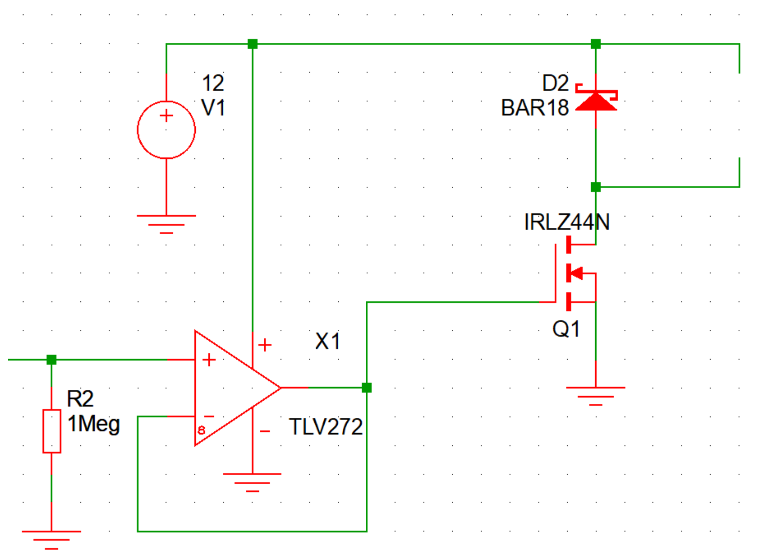
The 24V battery is a FLEX brand lithium ion drill battery. This battery was chosen to allow for use of the brand's corded variety, and initially it was determined that we would make use of the stepper motor's 24V capabilities. Due to design complications, we decided to use our 12V delivery system to maximize time that we had with a working device for debugging prior to EXPO, because our initial 24V design was having power issues.

Overall software

We used FreeRTOS to create a responsive, modular, and real-time embedded control system. The code was organized around using tasks, interrupts, timers, and event flags, which allowed different modes to operate independently while reacting quickly to user input. At the core, there were three FreeRTOS tasks: TaskCastMode, TaskReelMode, and TaskEmergencyMode. Each task handled a specific mode of the fishing device, that were triggered by button presses. Event flags were used to signal these tasks, doing it this way creates asynchronous behavior, and not needing to resort to delay-heavy code or blocking code. We used interrupts on three buttons, along with periodic sampling of an analog lever-style joystick with a FreeRTOS timer. The joystick input controlled LEDs and motor speed or cast distance, giving the user intuitive control over casting distance and reeling speed. Overall, the software architecture balanced real-time responsiveness, modular task separation, and hardware safety.

Reel Motor Driver

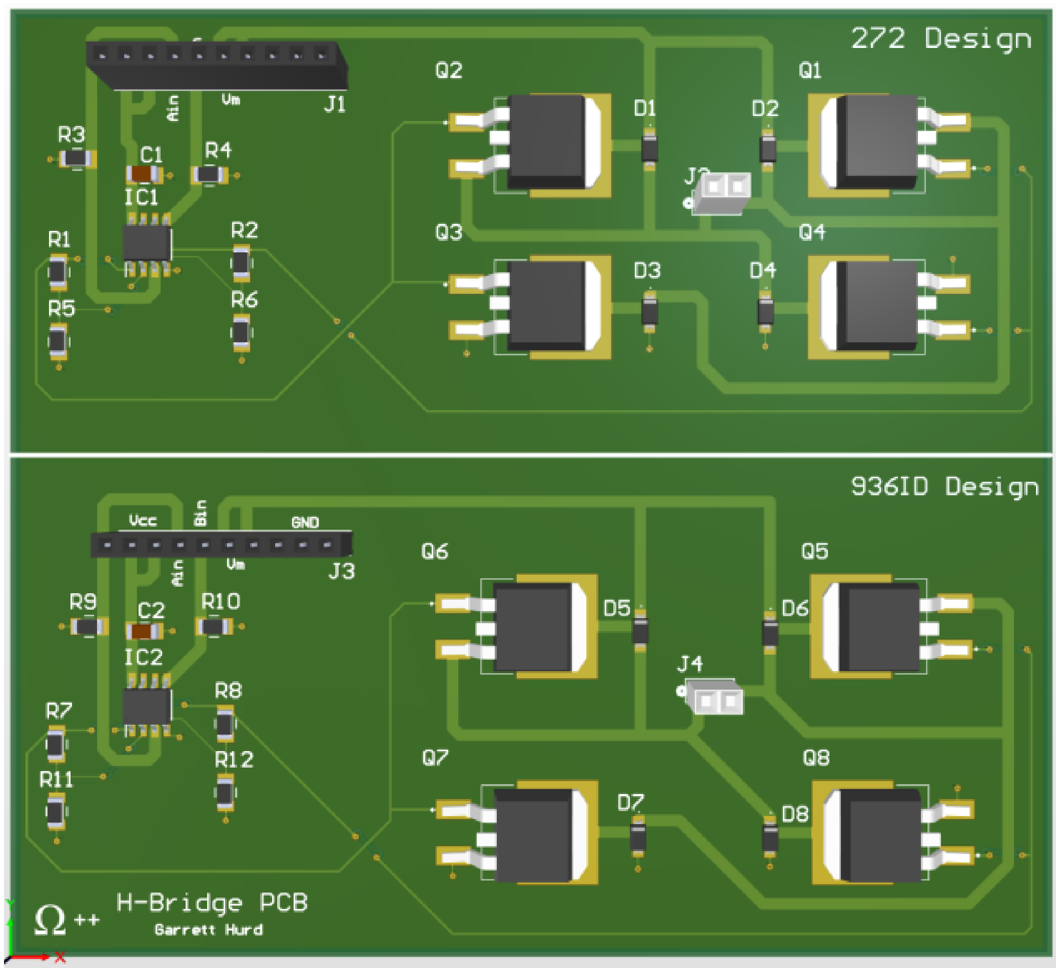
The reel motor used a low side mosfet and PWM to adjust the speed of the reel. This design was very simple but effective and used very little power. Originally, we were going to add a compensator to regulate the speed but decided against it since we wanted the reel speed to naturally decrease when the tension in the line goes up to prevent the line from snaping. This was one of the few parts of the project that went very well and ran into very little issues with the only setback being a few days of waisted time designing a speed compensator.



Reel Motor Driver Schematic

H-Bridge Design

The H-Bridge Design gave us many issues throughout the year but the design we eventually landed on has been a reliable and power efficient design. The design always included NMOS transistors and originally used a gate driver to pull the gate voltage well above the drain voltage. However, since we were unfamiliar with and didn't use an eval board for a gate driver we ran into several issues with debugging and reached a point that we either needed to pivot to a new design or double down on the gate driver. Since we weren't very confident in the gate driver design, we decided to transition to a logic control system using an opamp as a gain buffer. In the end, the only downside with this design was when a high voltage was applied to both terminals on the board, the battery would short through the mosfets and cause an incredible amount of power dissipation. This issue was solved via a buffer in the switching which helped solve the issue. The only other issue we had was with the back EMF from the spring moving the stepper motor. To fix this problem, we added Schottky diodes in parallel with the mosfets to better dissipate the current. These diodes proved to be extremely important as without them, the return of the rod to normal by the spring was not smooth resulting in a stuttering motion. This design was used for both the stepper and reel motor to save on design time. Before submitting the final board, we used smaller boards to test 2 different designs with different opamps to use a high gate voltage. We ultimately didn't increase the gate voltage since we wanted to fully test this idea but didn't have the time due to debugging.



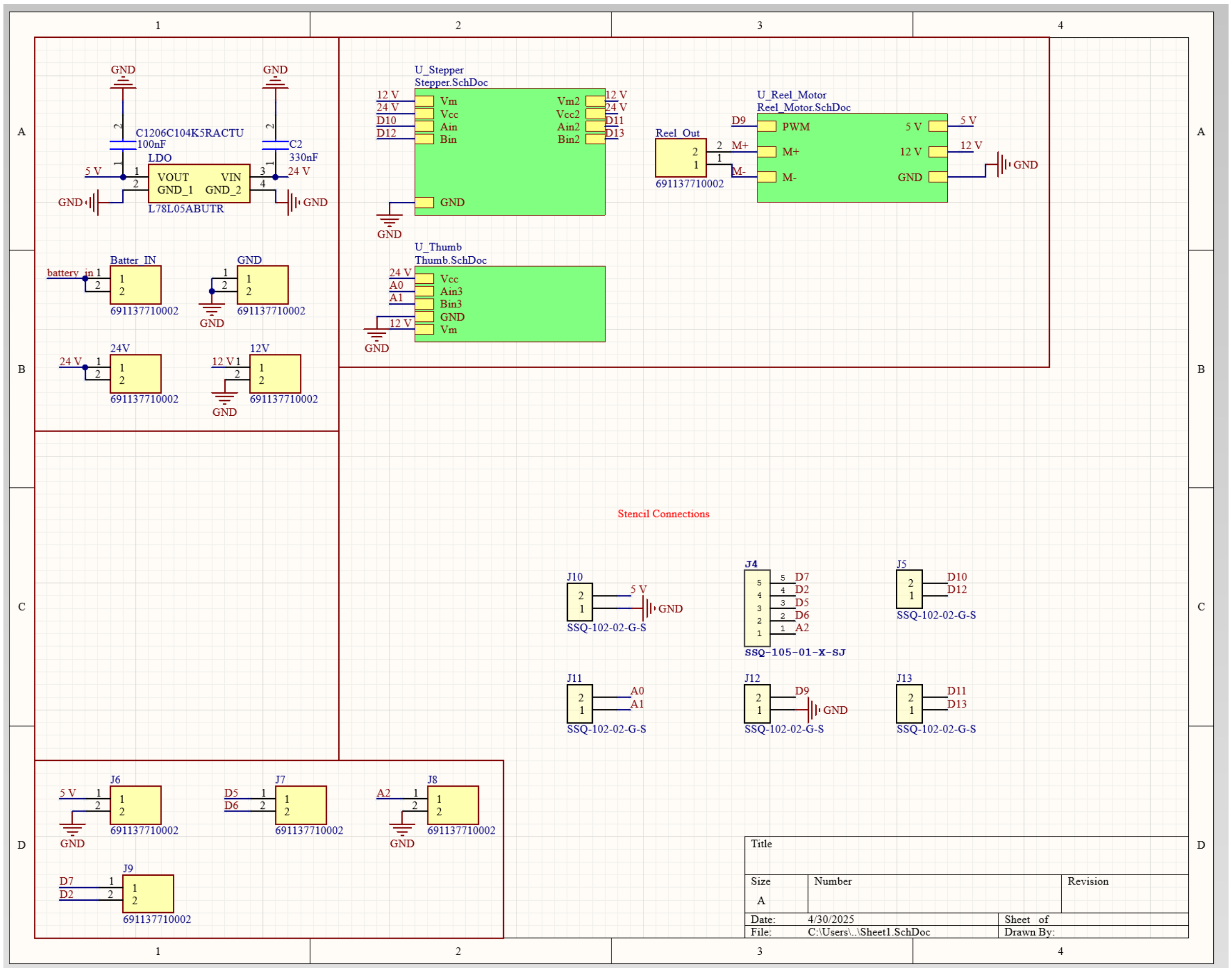
H-Bridge PCB

PCBs:

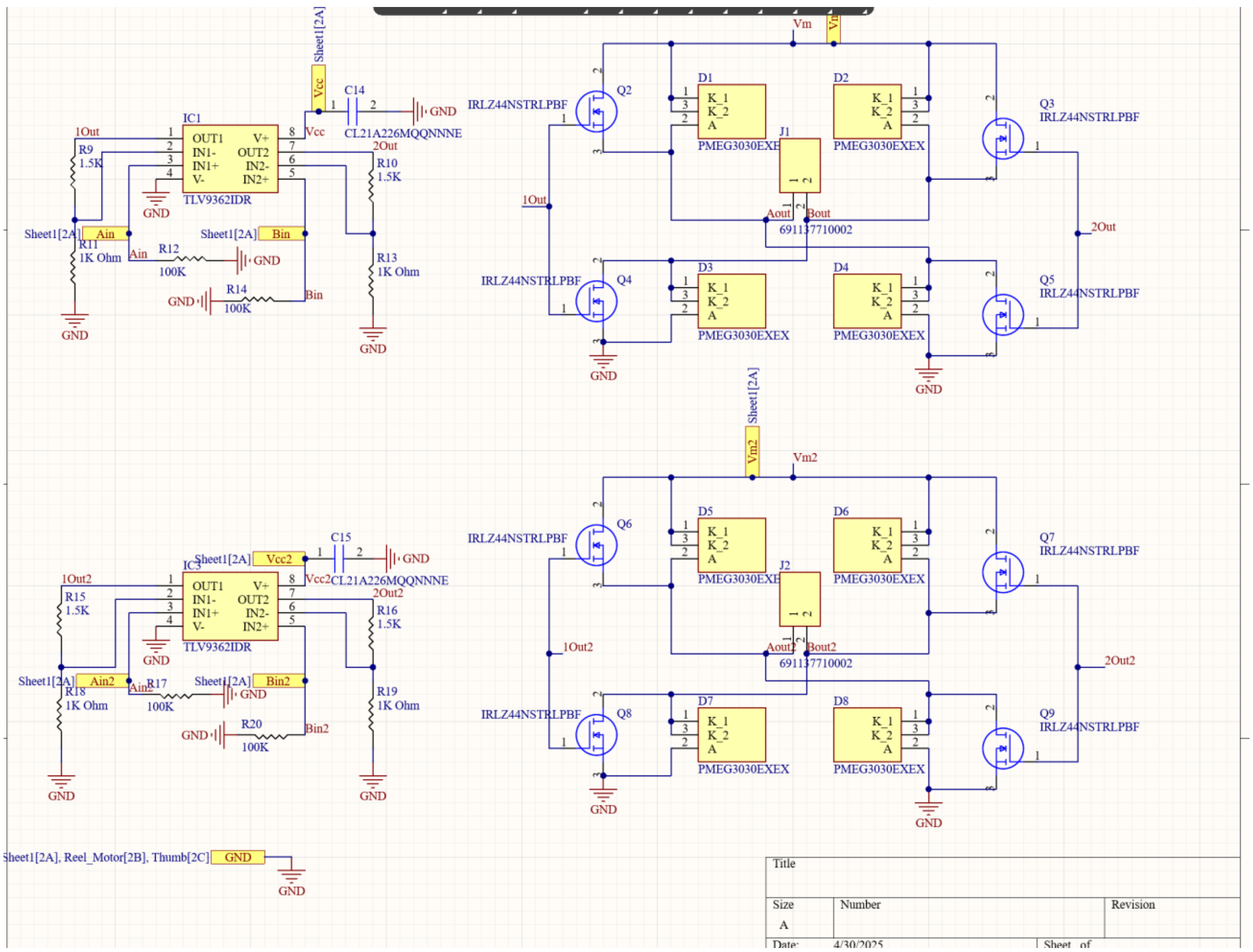
Our final boards consisted of two PCBs that would stack and attach on top of one another as shown in figure 1. The reason why I didn't just make one big board was due to size constraints of the housing this was going to be kept in. I also didn't try to fit in all one board by making it a 4 layer board because of two reasons. While I know how to make and route 4 layer boards I didn't want to risk trying something I haven't done before on our final board, especially since we were very short on time. The second reason why I didn't want to make a 4 layer board is because I still wanted a little bit of freedom to replace anything that wasn't working properly,for example the buck converter.

Picture of a backup board

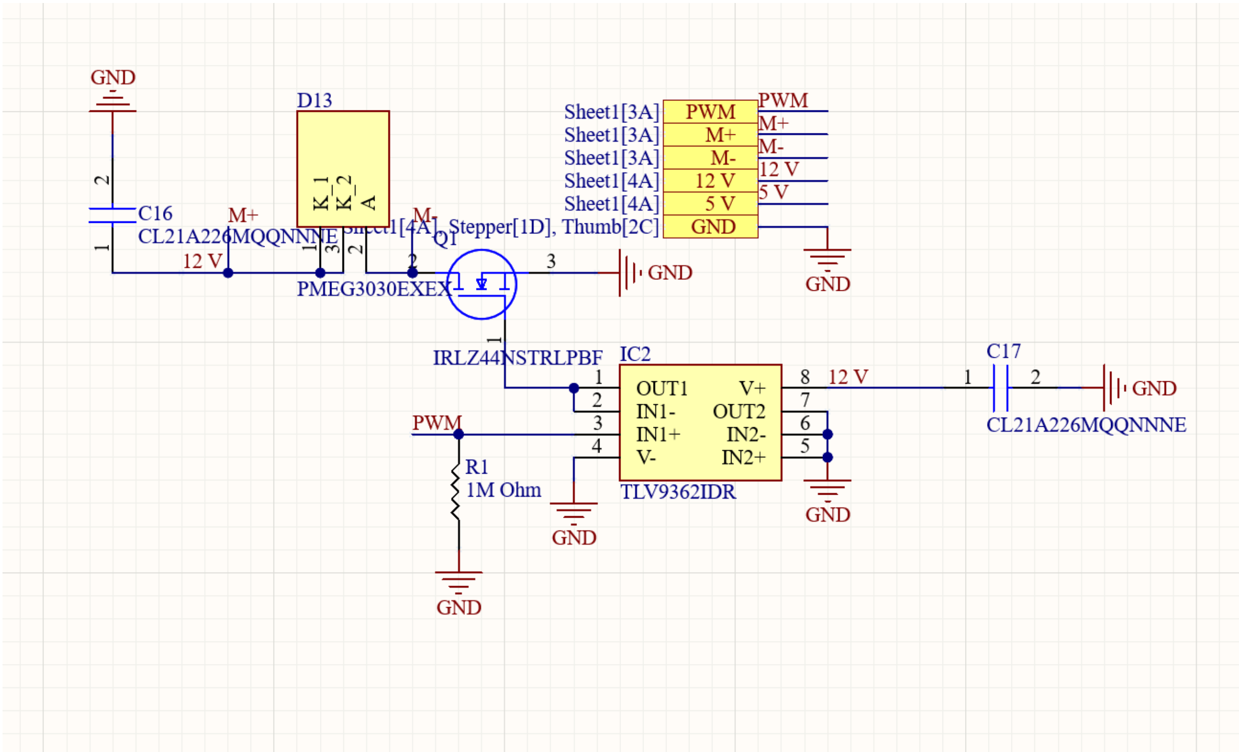
The bottom board consisted of screw-on terminals for the power connections, controller I/O connections, and motor driver outputs, a 24V to 5V LDO, and motor driver circuits for the DC motor, the stepper motor, and the linear actuator. Both boards include mounting holes so they can be mounted in the housing and to each other. Even though the original design was intended to be used with 24V, I left a few extra screws on terminals so that if we needed to change the voltage going into the drivers to 12V, which we did, we would just have to re-arrange some wires.



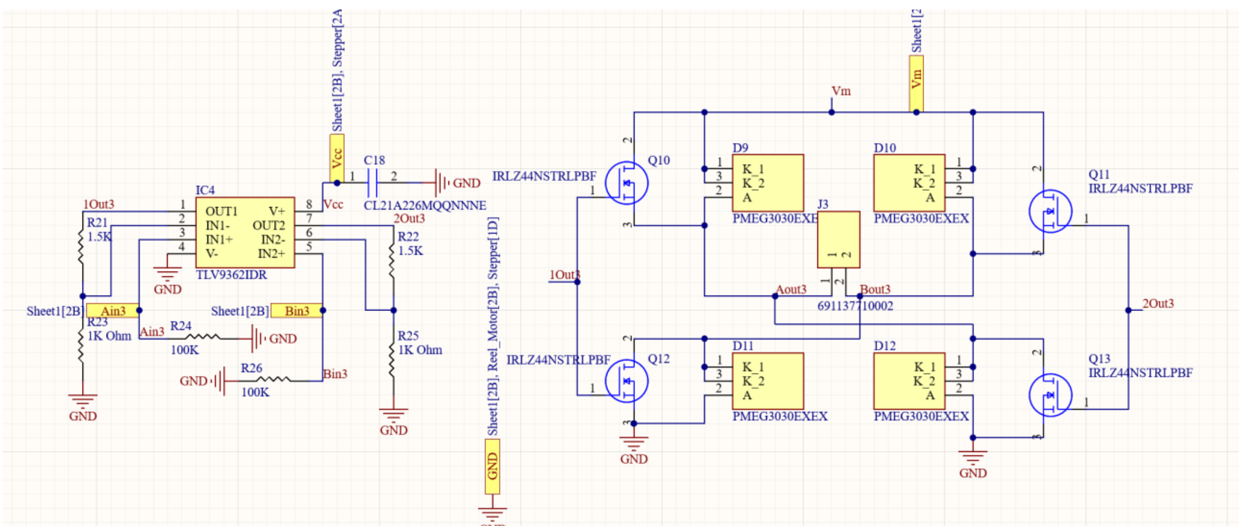
Altium schematic of bottom board



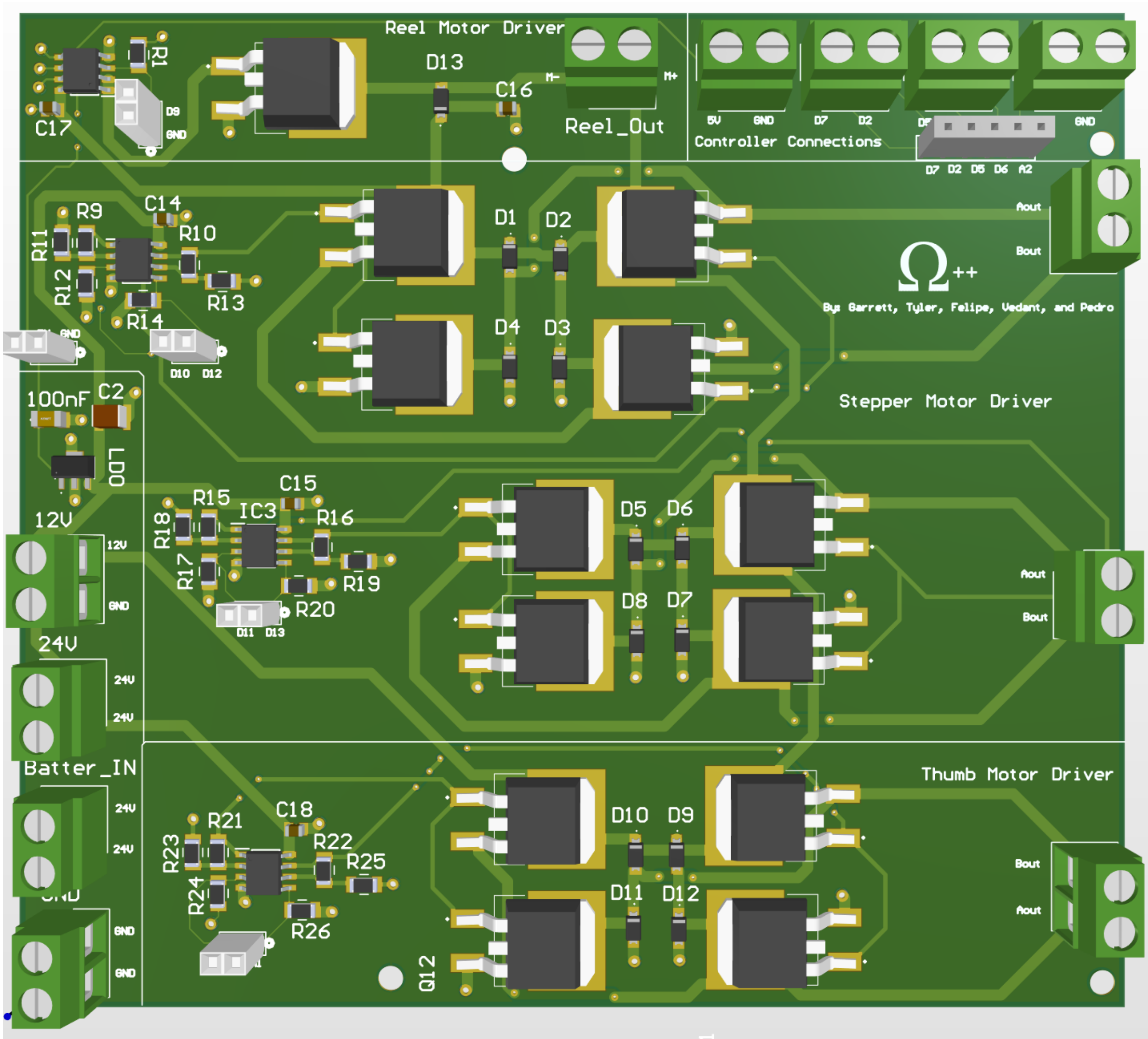
Altium schematic of stepper motor driver



Altium schematic of DC motor driver

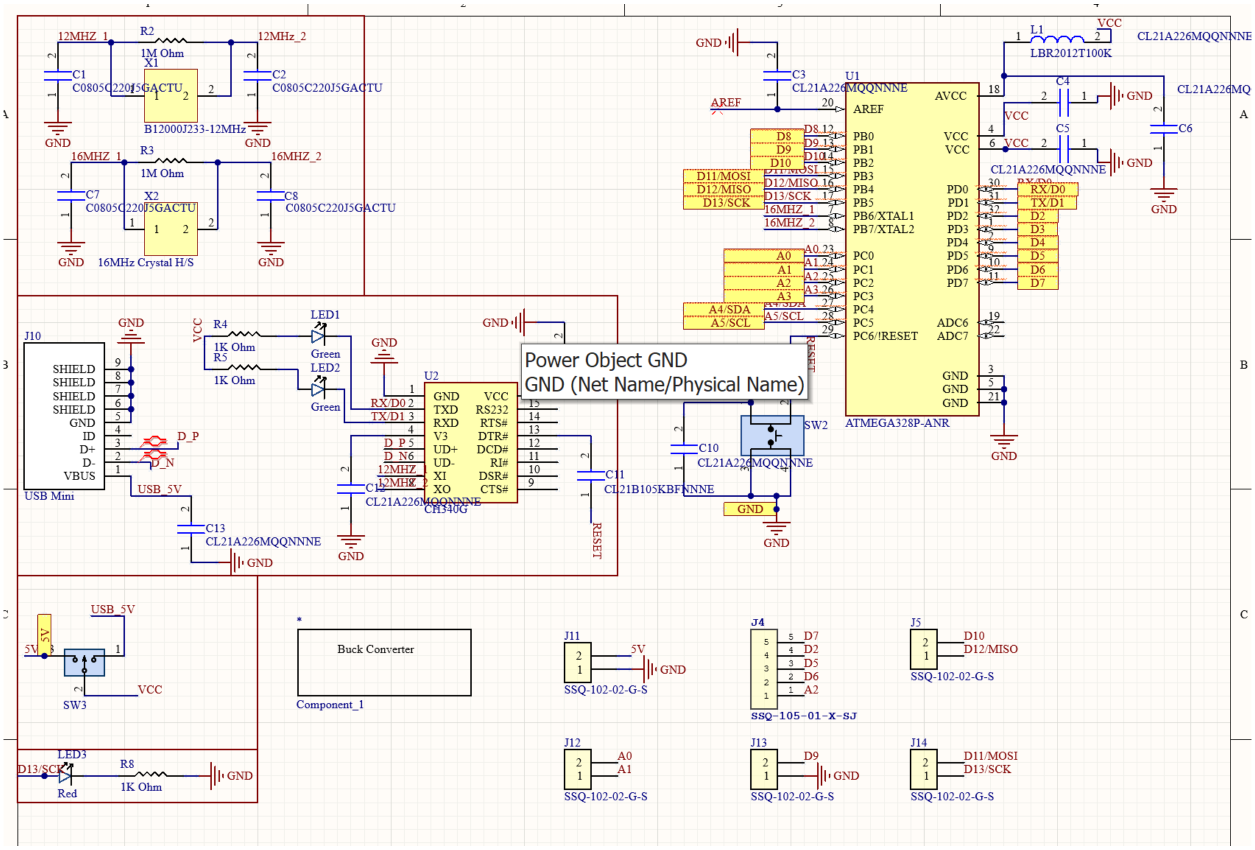


Altium schematic of linear actuator driver

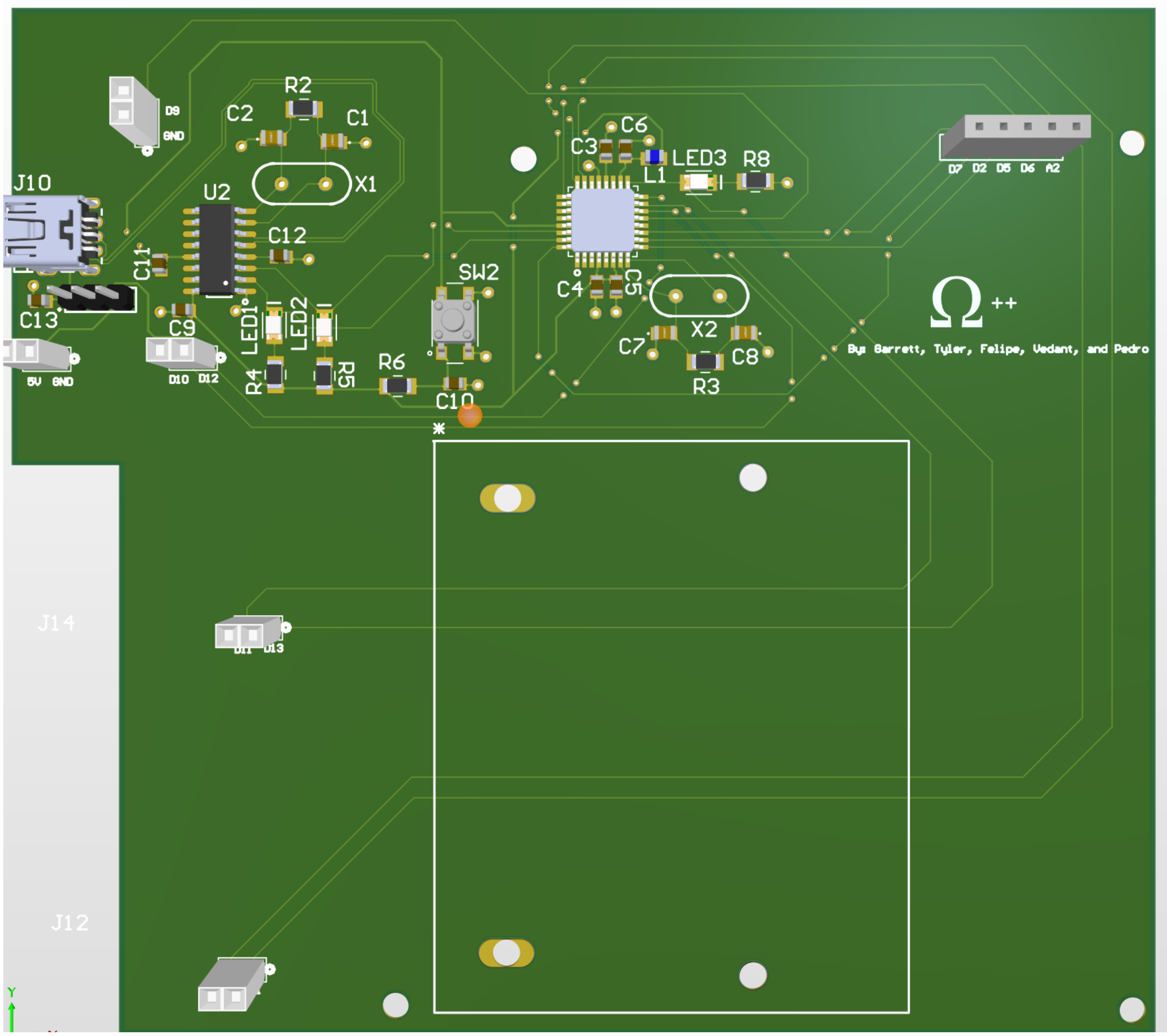


3D image of bottom PCB The top board consists of a custom arduino and a buck converter.

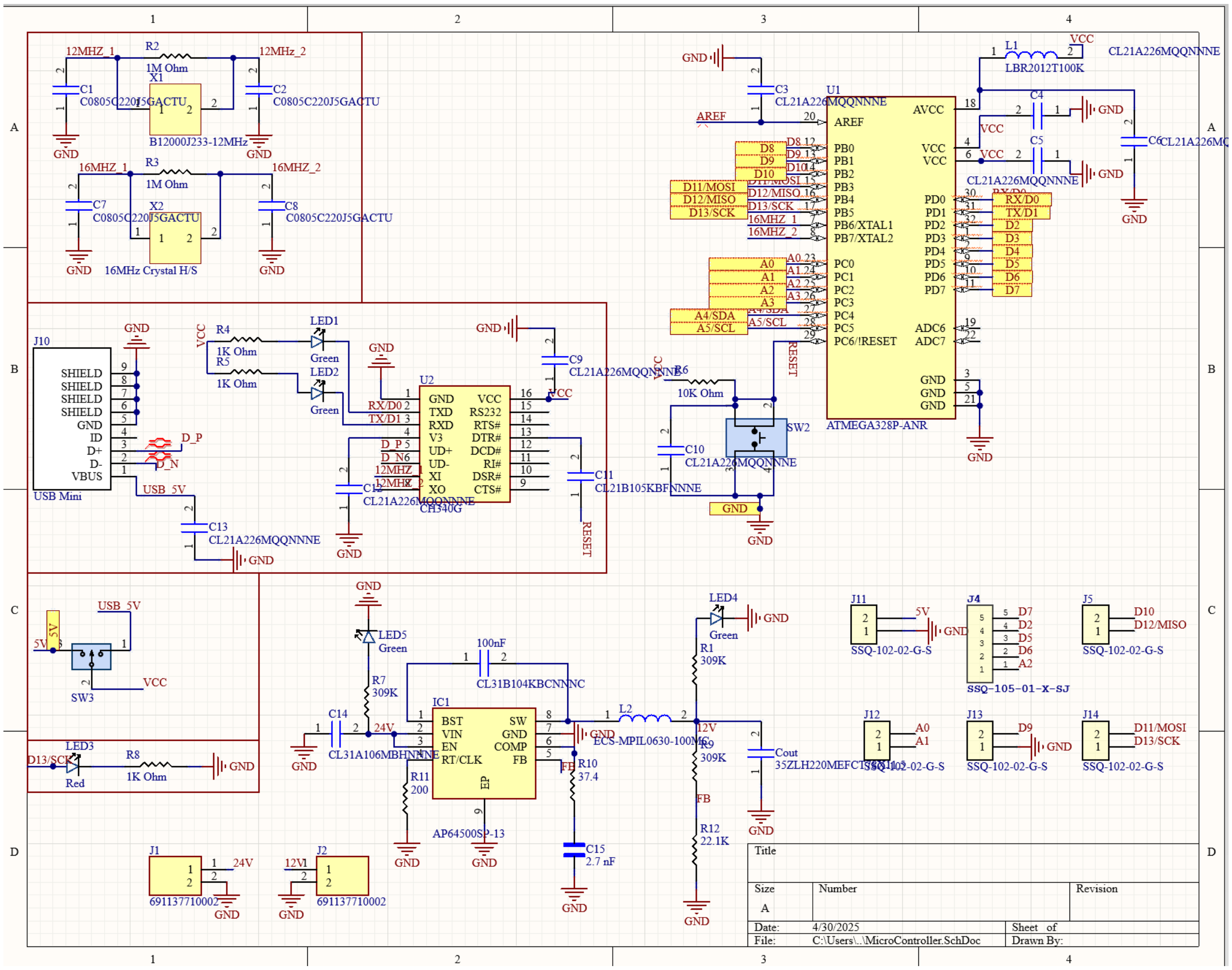
We originally planned to use a buck converter that would step the voltage down from 24V to 12V. When I initially tested my custom buck converter it seemed to have worked fine, but once we tried to incorporate it into the rest of the system during acceptance tests it immediately fried our buck converter. I designed the PCBs beforehand and since the buck converter was one of the highest risk components of our project, I made sure to incorporate a way to incorporate our back buck converter should our custom one fail, which it did. I made two versions of the top board, one with the custom buck converter and the other with the commercial buck converter. As mentioned before the top PCB also has the same mounting holes as the bottom board, it also has header pins to connect any digital pins from the arduino straight into where they need to be for the bottom board. I messed up 2/4 of the mounting holes for the custom buck converter but the buck converter seemed to hang well enough with the two holes I didn't mess up.



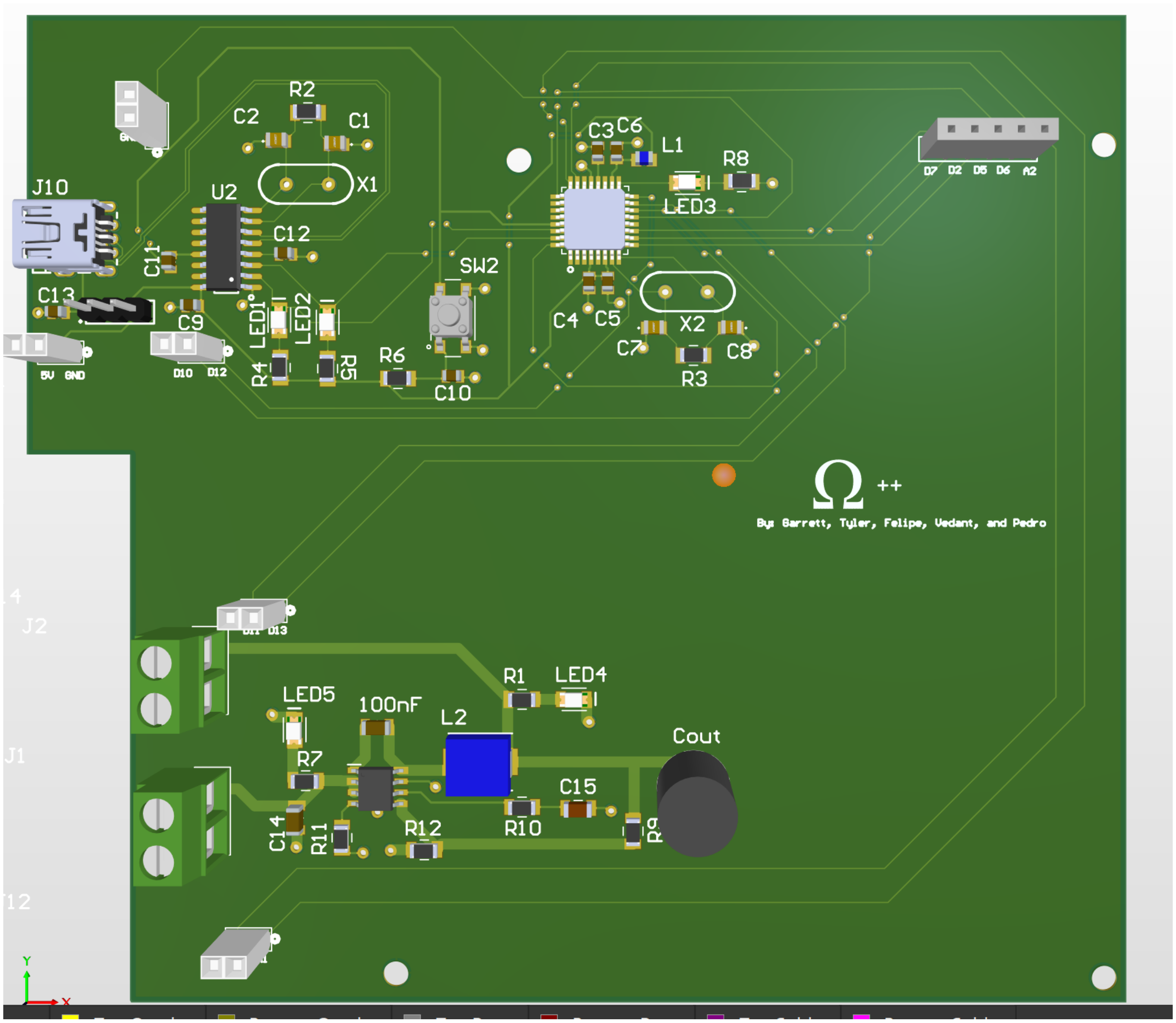
Top board schematic with commercial buck converter



Top PCB, commercial buck converter



Top board schematic with custom buck converter Figure 10: Top PCB, custom buck converter



Top PCB, custom buck converter

Controller and Electrical Enclosure:

The controller only has the control lever, the reel mode and cast mode buttons, as well as the emergency stop button. We wanted to maintain as much of the electrical system in the housing as possible, which enables our sponsor to make spare controllers for a cheaper cost, and easier manufacturing due to minimal parts. This decision was based around our sponsors informing us that our end users were likely to drop the device multiple times. This also led to the decision to use metal buttons, because they have a more durable material.

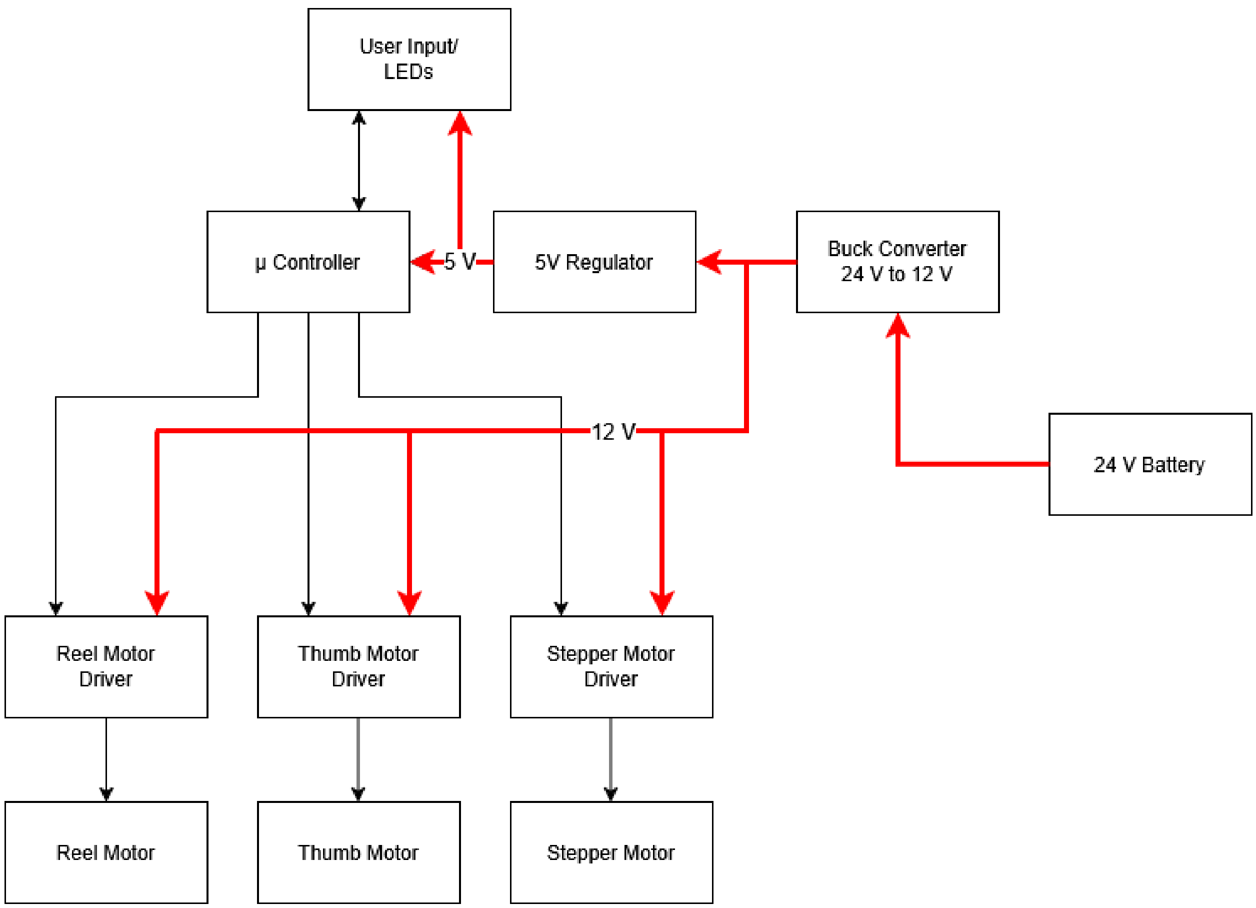
This durability design framework led us to coating both the controller and the enclosure in truck bed lining. This is a non-slick rubber surface that is designed to tolerate furniture being slid across it, gravel being slammed down onto it, and any other amount of torment that befalls a heavily used truck bed. This doubles as a water barrier, normally intended to prevent rust but in this case intended to prevent water intrusion during a rainy day due to our device housings being made with a 3D printer. The material chosen was ABS due to the ability to seal it with acetone, preventing small intrusion paths between print layers, but also it turned out the ABS softening temperature is significantly higher than PLA. With ABS softening at 221F compared to PLA softening at 140F, we feel confident that the materials will survive being left in the back of a van on a hot July day.

The layout of the controller has the minimal operational pieces, focusing on just setting a mode for the lever and an emergency stop. The lever is mounted on the right hand side to prevent users from having to reach across their body to operate the controller during extended fishing trips. The buttons are large in size, and elevated so that they may be operated by only using a wrist or fist.

Product Operation

"The How"

This device takes user input from our controller; those inputs go to our custom microcontroller that controls individual motor driver systems to imitate the user fishing. The device is powered by a 24V battery, this input voltage is stepped down by a buck converter to 12V to power our motor driver systems. We have a voltage regulator that steps down 12V to 5V to power our microcontroller and controller.



The reel motor and driver operate a DC motor that is attached to a modularly installed commercial fishing rod, nominally a Kast King brand bait caster due to the motor hex rod. This spins the reel, acting as the hand crank lever, and taking advantage of the design where you can shift the lever over to the other side to accommodate left handed anglers. Instead of shifting, it is removed and the reel motor shaft is installed

The Stepper Motor and associated driver control the casting action, by utilizing the stepper motor steps to apply torque to the upper portion of the device, tensioning the axial springs and providing a spring potential energy. When the desired back-cast distance is achieved by the stepper motor, the driver is secured and the motor is enabled to free-wheel in the opposite direction, casting the bait and tackle forward.

A thumb button driver controls the push-button on a bait casting reel, pressing it to lock the fiction clutch in place, which enables the line to be stationary during the casting motion. The software has timing set for the ten different cast distances associated with the Cast Distance LEDs that increment during the phase of setting cast distance. When the rod is in the forward-cast motion, the linear actuator is returned to the neutral position at such a time that the bait and tackle are propelled forward at a 45 degree angle, by releasing just prior to the apex of the arc. This timing was determined by repeat testing using a slow-motion camera to determine the exact release point given the linear actuator releasing the thumb button.

Product Analysis

G: One of our sponsors' requirements for this project was for the device to last 2 hours on a signal charge. The battery we decided to go with was perfect for this requirement. We calculated that if the whole system was drawing current all the time, the 12 Ah battery would last for 2 and half hours. However, when the system is ideal there is little to no current draw. The stepper motor would draw around 4 amps of current when in use and the reel motor would draw around 1 amp when in use. Since these motors aren't always being used, the battery would last longer than 2 and half hours. We confirmed this during Senior Expo. Our battery lasted the whole event and still had around 75% of charge left!

One design issue we ran into was our buck converter. We wanted to create a custom buck converter for our power system. We realized during phase 2 that we didn't account that we needed to add a feedback loop to our buck converter design. We decided to go for an IC buck converter that had a feedback loop inside of it as a Plan B. Sadly, this IC buck converter didn't work, and we didn't have time to design another one before Expo. We decided to go with our plan C, an off-the-shelf buck converter. We got this buck converter working before Expo and powered the system efficiently.

TM: The biggest frustration of what didn't work was the gate drivers. We took a risk with a technology we were less familiar with which didn't pan out. We ultimately switched to a new design that we were familiar with, but this decision ate away about a month worth of time already spent and was a large source of grief for the group. When we made the switch, it was a tough decision but we're very happy we did because at the time, there was no certainty that the old design would work. The discussions that the group made was an important and difficult one that ended up paying out for us but took a lot of consideration or the risk of both options.

F: Our final design successfully allowed users to operate the fishing device with a single adaptive controller, which fulfilled the primary goals outlined in our MRD and ERD. One of our initial goals was to have our device support multiple types of input devices (e.g., bite tab) in order to accommodate a larger range of users' abilities. Our hardware was modular enough to allow for future expansion, but time constraints and running into integration issues made it so that we only implemented one controller type. This allowed us to ensure we delivered a functional MVP (Minimum Viable Product) within the time limit.

P: Most of the circuits we had planned for our final board had worked with the exception of the custom buck converter. We also seemed to have met all of our sponsor's requirements for the project. The project is able to fish for more than 2 hours, it is able to cast an acceptable distance, and our controller design was approved by our sponsors. If we had more time to work on the project the first thing we would do is work on integrating different controllers such as bite tabs and head switches into our design. We would also alter our design so that if the user pulled the switch back it would activate the stepper motor and pull the rod back. It will continue to go back until it either reaches the maximum point or if the user pushes the switch forward. This change would make the controller more intuitive.

TW: The battery power issue was something that we put a lot of effort into, and that we ended up overshooting by a mile. Our large battery that we had hoped would last for way longer than needed, ended up lasting longer than we would almost even want. We have a small fear that the battery will go through enough trips without needing a recharge that our sponsor may forget where the charger even is! Our initial design choices with the battery were done at the proof of concept phase, and we calculated at 30 casts/hr, that we would use 1.26 amp hours, where as we used the 2.5Ah battery for mini-EXPO, and the device was constantly running for a few hours, and we just barely got to the 75% indicator on the on-board battery power indicator.

Current Consumption Calculations

Ohm++

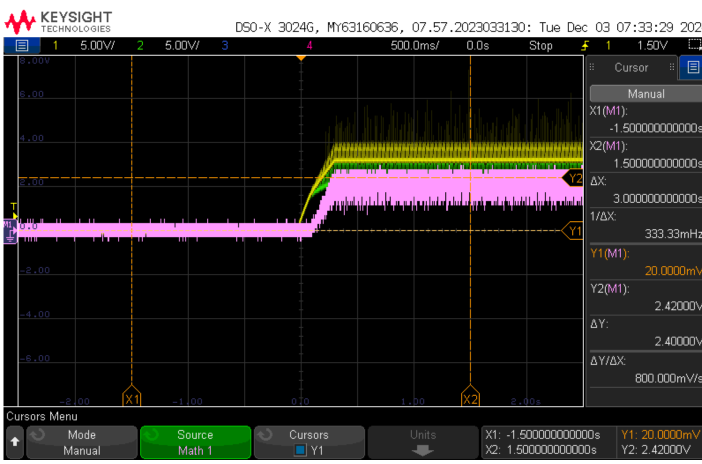
Adaptive Fishing Rod Controller

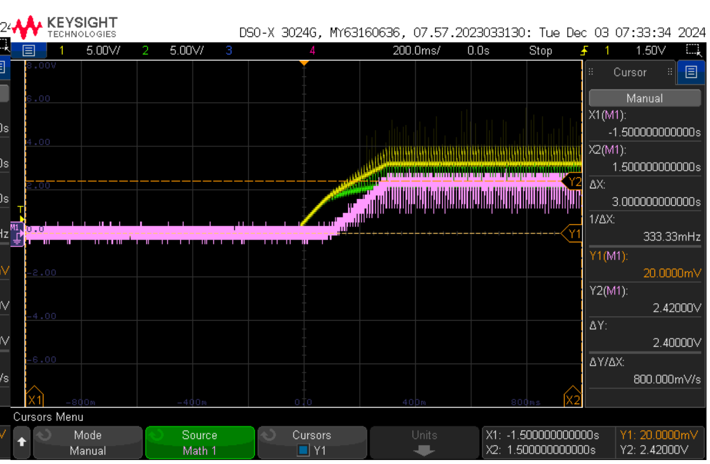
BLUF: Two hour fishing trip with 40 casts minimum,

2(0.16Ah + 0.0456Ah + 0.00072Ah) = 0.4126AH

Safety Margin: 30 Casts/Hr, constant reeling for entire hour = 1.26Ah for two hour trip.

<u>**Casting Motor**</u>





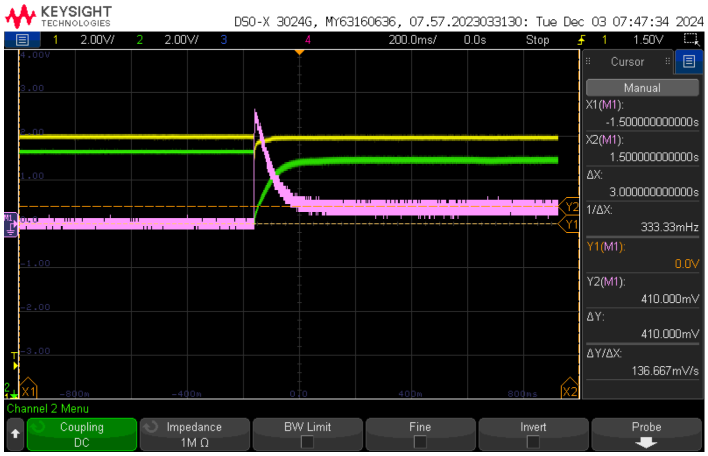
$$I = \frac{2.4V}{1\Omega} = 2.4A$$

$$Max\ Back\ Cast = 12 seconds$$

$$\left(\frac{20\ Casts}{Hr}\right)(12 seconds) = 240 seconds = 4 minutes = 1/15\ \text{Hr}$$

$$4 minutes\ @\ 2.4A => \frac{2.4A}{15} = 0.16AH$$

Reeling Motor



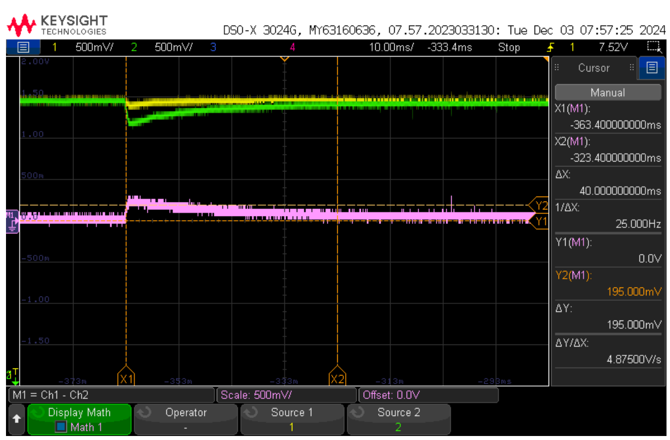
$$I = \frac{410mV}{1\Omega} = 0.41A$$

$$20ft\ Reel\ Time = 10\ seconds => 40ft\ {\sim}20\ seconds$$

$$\left(\frac{20\ Casts}{Hr}\right)(20seconds) = 400seconds = 6.66\ Minutes = 1/9\ \mathrm{Hr}$$

$$6.66minutes\ @\ 0.41A => \frac{0.41A}{9} = 0.0456AH$$

Thumb Button Motor



$$I = \frac{195mV}{1\Omega} = 0.195A$$

$$40\ Actuations\ for\ 20\ Casts\ @\ 333.3mS = 13.3seconds/Hr$$

$$13.3\ Seconds\ @\ 0.195A => \frac{0.195A}{270.67} = 0.00072AH$$

Lessons Learned

- Iterative design isn't only necessary, it enables you to smooth out a design and apply lessons learned from each of the previous iterations to end with a significantly better design than which you started with
- The ability to discard poor feedback is underrated. It requires you to assess input based on the assumption that it may be valuable, and compare it to the current design and come to a specific decision to discard that input. Other people will have things to say about your work, and you need to be equipped professionally with the proper tools to say "No."
- Through this project, I learned a lot about the integration of hardware and software, which was a completely new area for me and it also very helpful as that is how things are. It helped me realize how interconnected these two components are, and how even small issues on one side can cause the whole system to fail. One of the biggest takeaways was how important attention to detail is—sometimes the tiniest mistake, like a misplaced variable or a single wire not connected, can stop everything from working. I also improved a lot in debugging; I became more patient and methodical in tracing problems, and I started to better understand how to isolate and fix errors step-by-step.
- Learned how to effectively communicate with others technical information and adjust the way I speak to multiple audiences
- Improved on finding information and doing research on my own going from not knowing anything about stepper motors to designing a motor driver for one.
- Drastically improved my ability to make 'Engineering Decisions and justify why those decisions were made
- Learned to work with a team of six on a complex engineering project
- Used the skills and knowledge I gained from my undergraduate EE class and apply them to this project
- Understood the importance of customer requirements.
- Learned to create and manage deadlines during the development of the electrical systems of the project.
- Learned to manage the software and hardware teams on this project to meet deadlines and ensured that both teams were on the same page.
- We learned how to be a leader of an engineering team
- We learned how to manage my stress during the development phase of the project

- We gained confidence in public speaking, leadership and being an engineer.
- Making custom footprints in Altium
- Importing footprints from internet
- Plan for the worst case scenario, examples: PCB doesn’t arrive until 4 weeks after its order or testing/troubleshooting will require more than 1 week
- Don’t order bare minimum amount of parts, buy plenty of extras since components can easily be lost or broken
- Reach out for help when it's needed, don’t spend too much time trying to work on a task alone when it could be done faster with a teammate.
- Team collaboration and communication, coordinating between hardware and firmware, taught me the importance of clear communication, version control, and being flexible when integrating subsystems.

A Year In Review

Team Reflections

Felipe:

During my time in capstone, I grew both technically and personally while collaborating with a diverse and talented team. My primary role was focused on embedded systems and software development, more specifically, implementing the control logic for the adaptive fishing device. This gave me hands-on experience with real-time systems, hardware and software integration, and user interface design. A high point for me during capstone was seeing our device perform reliably during final testing. Looking back, I would have probably advocated for earlier prototyping and clearer interface definitions between software and hardware components. A lot of time was spent reworking portions of the system because of mismatched assumptions. Earlier communication would have saved effort and made integration smoother. In future complex projects, one thing I'll do is prioritize modular development to reduce dependencies.

Pedro:

I had a couple of takeaways from working on this project, from a technical standpoint I became more familiar with Altium, designing and importing footprints. From a logistical standpoint I learned that I should be better prepared for the worst case scenario, PCBs arriving late or testing/debugging taking longer than expected. I also learned that I shouldn't order the bare minimum amount of parts for my boards since components often get lost or broken.

From a personal standpoint I learned to be more open to help from my teammates instead of trying to finish all of my tasks on my own. Having at least one other person working on something exponentially decreases the amount of time that I would otherwise spend doing it on my own. This capstone project definitely had plenty of low points. One of the most infuriating problems I faced in this project is I would invest a substantial amount of time into a PCB and then after 3-4 weeks I realize I made a major mistake on the PCB. Then I'm left with the decision of should I still try to see if I can make the circuit work despite the mistake or should I move on to the backup option for the circuit since I invested so much time for no return at all. Another big low point for this project for me is when I couldn't get my first iteration of buck converter to work. After I couldn't make it work, I made a second iteration , this time using a Buck IC that seemed easy to use and understand. After I soldered it and tested it using the capstone power supply, it worked perfectly, but once I connected to the power supply of our project it instantly melted some of the pins of the Buck IC. As a result we never got to use a custom buck converter and we had to stick with our commercial buck instead.

Another big low for this project was when a week before we had to prepare for acceptance tests, all of our circuits worked perfectly fine with no apparent issues, but when we tried to shove all of our circuits into the housing of our project we ran into non stop issues. It was also very disheartening to see our project which worked perfectly fine a week ago suddenly start heating up and either melting wires or start melting solder joints. Even though we experienced many lows in this project we did experience a couple of highs. A big high for me is when we got all of our PCBs working and in the housing for mini expo.

I'm very accustomed to my projects failing spectacularly right before I have to present them, but I'm very glad this wasn't the case for mini expo and expo(sort of). I loved seeing our classmates mess around with our project and actually be able to play the little mini game we set up. During mini expo I would periodically check the project for any faults and I would actively look for things that could be wrong since it just seemed too good to be true. To my surprise our project far exceeded our expectations and had worked very well. I will admit there was a hiccup with the buck converter during the expo however we found that just by keeping the device on we were able to get past this problem. Luckily for us our battery lasted the entire expo event with people consistently casting it and after the event ended we saw that the battery still had at least 75% more power in it. If I could turn back time and do this project all over again there are a couple of things I would do differently. I would order our PCBs on our own since ordering through capstone was very unreliable and simply not acceptable when under a tight time budget. I would be less afraid of ordering extra parts for boards especially since components often either get lost or break. I would also have my teammates and TAs do a sanity check on my schematic/PCB before I order it.

I found by doing this it drastically decreases the amount of major mistakes that could end up in the PCB. For example, while I was finishing up my final board, I sent everyone in my group a link to my schematic and PCB. My teammates were able to point out major flaws in my design that I either didn't consider or overlooked. Having a fresh pair of eyes is very useful and can save me from potentially having to redesign a new board. I also plan to be more considerate with my choice of components and make absolutely sure that every specification is met with a component. For example, in the final board I used 0805 decoupling capacitors, however I realized I missed the fact that the capacitors only had a voltage rating of around 6 V. Since many of these capacitors were on 12 to 24V lines I had to solder 1206 22uF capacitors that could handle the voltage on an 0805 footprint. If I absolutely made sure requirements such as voltage rating were met I could have saved myself a sizable amount of time soldering. Based on my experience working in capstone I do have a couple of ideas of how I will approach future team projects.

As mentioned before, I plan to be more considerate with how I plan the progression of a project. I need to account for worst case scenarios where either something gets delayed or if a simple problem turns into a more complex one requiring more time. I also plan to write out a test procedure of what I need to test for a PCB before I order it. It can be difficult to recall what I need to test my design and it's just good practice in general to have a testing procedure already planned out. I would also make sure to run everything through my team before deciding any major decisions such as ordering a PCB.

Garrett:

From this capstone experience, I learned how to be an effective team leader for an engineering project of this scale and with a big team, learned how to communicate better between teammates, learned the importance of meeting deadlines/scheduling, learned how to take reasonability/accountability as a leader, and ultimately, I gained confidence in myself and as an electrical engineer.

One of the lower points of this experience was when one of our experimental stepper motor drivers wasn't working as intended. We attempted to fix the issue through debugging for a week, ultimately, I made the decision to move on and go forward with our plan B since we didn't have the time to debug any further.

The highest point during this experience was during Capstone Expo. Our system and controller were working as intended and everyone who was using it was having fun using it. Our sponsor showed up and was extremely happy with the results of our development. This moment was awesome, all the time and effort my team and I put into this project finally paid off.

If I was to do this project all over again, the major thing I would do more effectively would be scheduling of deadlines. Most notably, I would use a Gantt chart. Secondly, I would encourage more team collaboration. There were points during this project where team members would work independently, this caused some slight development issues and fatigue.

The experience I gained during this project and from capstone I will take with me into my professional career. All of it was extremely valuable and I am proud and honored that I was able to be a part of the Adaptive Fishing Rod Controller project, Ohm ++ and CU Boulder's Electrical Engineering Capstone Class of 2025.

Tyler:

Throughout the course of the year, I have felt a whole range of emotions with this project. Frustration, annoyance, relief and joy. It has definitely been one of the most unique experiences I've had in all of college. It felt so different compared to any other course I've taken using engineering judgements to make decisions instead of learning math or theory. I really appreciated this experience and seeing the final project was incredible. I really felt like an engineer and after all the time spent in lab, seeing the fruits of my labor brought the biggest smile to my face. Despite all the stress, late nights, and debugging, I'm extremely happy and proud of the work I put forward.

Vedant:

Overall, I feel that the capstone was a great experience. It gave me the opportunity to work with people I hadn't met before, and despite that, everything came together really well. Each team member was confident and capable in their own tasks, which made the collaboration smooth and efficient. We had very few disagreements, and even those were resolved quickly and respectfully. It was great to be part of a team where everyone was focused and passionate, and I think that played a big role in how successful our project turned out to be. Some lows we had were the system completely stopped working before demos and to debug that just for a few mins was intense but we were always able to solve it and got our project working just what our sponsors wanted which was the biggest high. One thing I would do differently to our fishing controller is to have it cast in different directions through our controller and nor manually move it.

Travis:

This project was reminiscent of my time in the shipyard. The initial weeks feel insurmountable, and that feeling never goes away. You just look up one day and you're months down the road and a mountain of progress has been made. It's kind of like looking back down the mountain half-way through a hike, it's amazing how far you can go when you just put one foot in front of the other. Yes, some areas are steeper than others, but a little bit of progress goes a long way when daisy-chained together over a long period of time.

My personal progress felt a little off for me. I wasn't in charge of the deepest trenches of electrical engineering for this project, but my roles were nonetheless important. I spent a lot of time reading articles, watching videos, and otherwise contemplating UI/UX for people with spinal cord injuries. This paired with my previous leadership roles outside of school helped keep our team going in the right direction and while mitigating pitfalls along the way. While I absolutely learned a lot during capstone, I think that my biggest hope is that I had a positive impact on the members of our team. There are lessons that I have learned along the way that I learned the hard way that were seemingly learned with minimal pain over the course of the project.

I don't know that I would do much differently if we were to start over, or go back in time. I will say that I would jump at the opportunity to continue working on this project with my current team, and start accomplishing our initial stretch goals of a bite tab or head switch operated controller.